1 Waves - A primer

Learning Objectives

- Be able to define a wave and some parameters (phase, wavelength, wavenumber, period, frequency) describing waves

The terms *wave* conjures up many image – ripples in ponds, ocean swells, sound waves, a wave in a stadium created by the spectators and so on – essentially, something that involves an oscillatory motion. Formally, a wave refers to a traveling disturbance that accomplishes the transfer of energy or information. Some waves require a medium (e.g., atmospheric waves) and some do not (e.g., electromagnetic waves). A unique aspect of waves is that although the energy (or information) travels a great distance, the relative distance moved by the individual particles of the medium is minuscule. In other words, there is no net transport of mass (N.B. there are exceptions to this simple constraint, especially during nonlinear wave-wave and wave-mean flow interactions and in presence of viscous effects).

1.1 Wave parameters

To describe a typical wave, we need some parameters. These include:

1. Wavelength (λ): the physical spacing between successive troughs/ridges. A related quantity, wavenumber is defined as $k = 2\pi/\lambda$.
2. Period (T): the time elapsed between the passage of successive troughs/ridges.A related quantity, frequency is defined as $\omega = 2\pi/T$.
3. Amplitude (A): the maximum deviation in a physical quantity due to the wave passage.

A typical mathematical description of a wave that we will use in this course is written as follows:

$$\psi(x, y, t) = Re\{Ae^{i(kx+ly-\omega t)}\} \tag{1}$$

Where $i = \sqrt{(-1)}$ and $Re\{\}$ represents the real part of the quantity in the curly brackets. How does this represent a wave? Recall that Euler's identity can be written as:

$$e^{i\theta} = Cos(\theta) + iSin(\theta) \tag{2}$$

Therefore,

$$Re\{e^{i\theta}\} = Cos(\theta) \tag{3}$$

which clearly represents a wave.

Now returning to eq. 1, we see that the field ψ is *wavy* along both x and y directions with wavenumbers k and l respectively, and with frequency (ω). The quantity $kx + ly - \omega t$ is called the *phase* of the wave and is denote by θ. The numerical value of the phase will determine whether we are in the trough or ridge or somewhere in between. According to eq 1 , the phase is a function of both time and space.

The generic wave described by eq 1 takes on specific structures depending on the type of wave we analyze. Usually, for each wave type, we can derive a relationship between the frequency and wavenumbers. This relationship is called the *dispersion relationship*.Among other things, the dispersion relationship tells us what values of k, l and ω are allowed for any wave type.

1.2 wavepacket

When waves occur in nature, they are rarely of a single frequency and wavenumber. Instead, waves of a spectrum of frequencies and wavenumbers are excited. As an example, think of the sound made when you thump a table. The sound produced has multiple frequencies.

A wavepacket is a collection of waves with nearly the same wave parameters. When we add two or more such waves, the combined wave structure is called a wavepacket. For instance, lets add two sine waves that have wavenumbers and frequencies that are slightly off from the values k and ω respectively:

$$\psi_1 = sin([k + \delta k]x - [\omega + \delta\omega]t) \tag{4}$$
$$\psi_2 = sin([k - \delta k]x - [\omega - \delta\omega]t) \tag{5}$$

Thus, the combined wave is given by:

$$\psi_1 + \psi_2 = sin([k + \delta k]x - [\omega + \delta\omega]t) + sin([k - \delta k]x - [\omega - \delta\omega]t) = 2sin(kx - \omega t)cos(\delta kx - \delta\omega t) \tag{6}$$

where we have made use of the trigonometric identity:
$sin(A) + sin(B) = 2sin((A + B)/2)cos((A - B)/2)$.
The resulting disturbance consists of a product of two terms. A sinusoidal term that describes a wave with parameters k and ω, and a slowly varying envelope defined by the cosine term. It is slowly varying because of the small wavenumber (δk) and small frequency ($\delta\omega$).

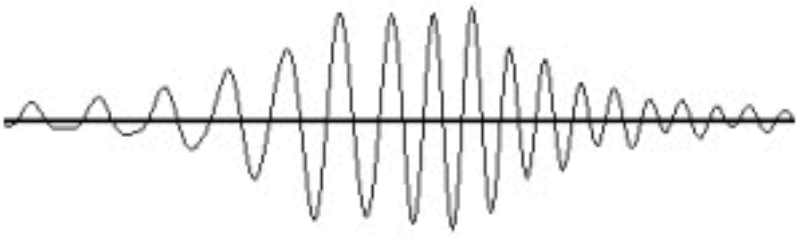


Figure 1: A wavepacket Source: www.ipod.org.uk

Thus, the combined wave appears like a localized wave, which we call a wavepacket. Physically, the appearance of a packet is produced due to reinforcement of the waves where they individual waves are in phase and cancellation where they are in opposite phases. Waves excited in nature are usually found in bunches or wavepackets since whatever mechanism forces them, it usually produces a spectrum of waves and not a monochromatic wave.

1.3 Phase speed

The motion of the wave can be described by *phase speed*. If we focus in a particular phase of a wave, say its trough, then the phase speed is the speed at which the wave trough moves. The phase speed is mathematically described as the ratio of the frequency and wavenumber.
Thus, we say:

$$c_p = \frac{\omega}{k} = \frac{\lambda}{T} \tag{7}$$

The above expression is likely familiar to you from physics. You may have applied the formula (esp. the one involving the ratio of wavelength λ and period T) to the speed of sound waves.

Atmospheric waves can propagate in zonal, meridional and vertical directions. So, we can define the speeds in those directions explicitly. For a 2-D wave, the zonal and meridional phase speeds are given by:

$$c_{px} = \frac{\omega}{k} \tag{8}$$

$$c_{py} = \frac{\omega}{l} \tag{9}$$